

LEAN SIX SIGMA | RETAIL BANKING ANALYTICS

Demonstrating CASA Acquisition through a DMAIC Analytics Framework

CASA lead conversion baseline, factor analysis, machine-learning lead prioritization and process monitoring

Integrated Project Report

Simulated-data method demonstration · Define – Measure – Analyze – Improve – Control

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Report date	30 September 2026
Methodology	Lean Six Sigma DMAIC (Green Belt-level roadmap)
Data	Synthetic retail-banking workbook: 1,000 customers and leads, 30 Relationship Managers, 10 branches, Jan–Mar 2026
Tools	Python (pandas, scikit-learn, XGBoost, LightGBM), R (dplyr, car, readxl), Power BI Desktop, Excel
Evidence class	Method demonstration and decision-support prototype. Not evidence of real-bank behavior, cause-and-effect, ROI or actual improvement.

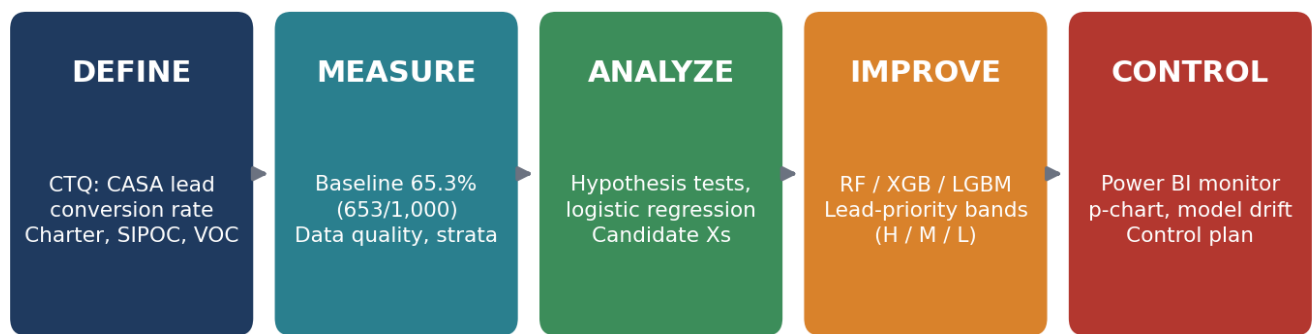
1. Executive Summary

1.1 Problem and approach

A retail bank has many CASA (Current Account and Savings Account) leads, but RM time is limited and there is no clear, repeatable process for deciding which leads to contact first. This project applies the Lean Six Sigma DMAIC roadmap to a simulated banking dataset to answer four questions: (1) what is the current CASA lead-conversion performance, (2) which factors are statistically related to conversion, (3) which leads should RMs prioritise, and (4) how can management monitor the process continuously.

The main Critical to Quality (CTQ) measure is the **CASA lead conversion rate**. The work moved from problem definition, through baseline measurement and statistical analysis, to a machine-learning lead-prioritization tool and a Power BI control dashboard supported by a written control plan.

Figure 1. DMAIC roadmap followed in this project and the principal deliverable of each phase.



1.2 Results at a glance

Phase	Question	Method	Key result
Define	What is the problem, the CTQ and the boundary?	Charter, VOC/VOB, CTQ, SIPOC, defect definition	CTQ = CASA lead conversion rate; formal sigma/DPMO deliberately deferred because no validated defect definition exists
Measure	What is the current performance?	Baseline, grouping, data-quality audit	653 / 1,000 = 65.3% (95% Wilson CI 62.3%–68.2%); monthly rates 62.9%–68.4% are not statistically different ($\chi^2=2.18$, $df=2$, $p=0.336$)
Analyze	Which Xs are associated with conversion?	Chi-square, Shapiro-Wilk, Wilcoxon, multivariable logistic regression, VIF	Supported: salary account (adj. OR 1.99), annual income, credit score, age, RM experience. Not supported: lead source, occupation, gender, RM identity. Model LR $\chi^2=93.4$ ($df=12$, $p<0.001$); McFadden $R^2=0.072$
Improve	Which leads should RMs prioritise?	Random Forest, XGBoost, LightGBM; randomized search, 5-fold CV	Tuned Random Forest selected on ROC-AUC: CV 0.672, held-out 0.604 (95% CI ≈ 0.52 – 0.68). Leads banded High 10.8% / Medium 59.5% / Low 29.7%. Used as an early decision-support tool
Control	How is performance maintained?	Power BI dashboard, control plan, p-chart design, model/data monitoring	One-page dashboard (KPI cards, trend, priority views, branch table) plus KPI ownership, SPC design and reaction plan

1.3 Main findings

a) Baseline:

1. We started with **1,000 leads**, and **653 converted**, giving a current CASA conversion rate of **65.3%**.
2. **“Two in three leads convert”**: means roughly **2 out of every 3 leads** become CASA customers.
3. **Data period**: We have data for only **3 months — January, February, and March**.
4. **Monthly conversion**: The monthly conversion rates were roughly between **63% and 68%**.
5. **No clear trend**: Because we have only **three months of data**, we cannot confidently say that the process is becoming **better or worse over time**.
6. **No unusual instability**: We did not find statistical evidence that the conversion process was behaving unusually from month to month.

b) Customer financial profile matters more than channel:

1. **Salary account matters**: Customers who already had a **salary account** converted at **72.3%**, compared with **58.8%** for customers without one.
2. **Odds ratio = 1.99**: After considering the other factors, customers with a salary account had about **2 times higher odds of conversion**.
3. **Income matters**: Converted customers had a higher median income:
Converted: **₹1,24,248**.
Not converted: **₹97,582**
4. **Credit score matters**: Converted customers also had higher median credit scores:
Converted: **729**
Not converted: **710**
5. **Channel did not matter significantly**: **Lead source, occupation, and gender** did not show a statistically significant relationship with conversion.

Predictive signal is real but modest.

1. **Predictive signal is modest**: The model has some ability to rank leads, but its AUC of **0.60** is limited. So, it should support RM decisions, not make automatic decisions.
2. **Control is designed, not proven**: The dashboard, SPC monitoring, and reaction plan are designed, but **not yet tested on real-time banking data**.

1.4 What this project shows and does not show

Evidence boundary

Demonstrates: a complete and consistent DMAIC analytics process; correct selection and explanation of statistical tests; a easy-to-understand ML prioritization tool that was tested honestly; and a control system.

Does not demonstrate: that the intervention raised real conversion, reduced defects, generated ROI, or achieved a long-term Six Sigma improvement. The data are synthetic, the observation window is three months, no before/after or controlled experiment exists, and statistical association is not causation. These claims would require a live pilot (Section 6.8).

2. Project Background, Objective and Data

2.1 Business context

CASA balances are a low-cost source of funding for a bank, so acquiring and activating current and savings accounts is an important business goal. Leads arrive from several sources (walk-in, digital, referral, campaign), are assigned to RMs, contacted and followed up, and either become CASA accounts or do not. Because RM time is limited, management needs to know not only how many leads convert, but also which leads should be contacted first.

2.2 Project objective

To develop a Lean Six Sigma-based retail-banking analytics and decision-support framework for improving CASA acquisition by

- (i) measuring the current conversion process,
- (ii) Identifying statistically significant factors associated with lead conversion
- (iii) prioritising leads using machine learning
- (iv) establishing a control mechanism for continuous monitoring of CASA, branch, campaign and lead-priority performance.

2.3 Research questions and hypotheses

#	Research question	Formal hypothesis (H_0)	Method
Q1	What is the baseline conversion rate?	— (estimation: $\hat{p}$ with confidence interval)	Proportion, Wilson CI
Q2a	Does lead source affect conversion?	Conversion is independent of source	Chi-square test
Q2b	Do income / credit score differ between converted and non-converted leads?	Equal distributions in both groups	Shapiro-Wilk, Wilcoxon rank-sum
Q2c	Are salary account, occupation, RM and campaign response associated with conversion?	Conversion is independent of each factor	Chi-square tests
Q2d	Which factors contribute jointly, after adjustment?	All regression coefficients = 0	Binary logistic regression, LR test, VIF
Q3	Can a model rank leads usefully?	Model AUC = 0.5 (no discrimination)	RF / XGBoost / LightGBM, ROC-AUC
Q4	Can performance be monitored and maintained?	—	Power BI, p-chart, control plan

2.4 Data description

The project uses a synthetic multi-table retail-banking workbook (Banking_Intelligence_Modified.xlsx) with nine sheets. The tables are linked to match the business flow: customers are linked to leads; leads to RMs; RMs to branches; converted leads to CASA accounts; accounts to transactions.

Table	Grain / rows	Key fields	Role
Customers	1 row per customer; 1,000	age, gender, occupation, annual_income, salary_account, credit_score, home_branch	Possible X (customer)
Leads	1 row per lead; 1,000	lead_id, customer_id, rm_id, source, converted (Y), lead_month	Outcome Y and process context
Relationship managers	1 row per RM; 30	branch_id, experience_years, monthly_target	Possible X (resource)
Branches	1 row per branch; 10	branch_name, city, monthly_target	Segmentation
CASA accounts	1 row per opened account	account_type, current_balance, branch_id	Post-conversion outcome
Transactions	1 row per transaction	account_id, txn_type, amount	Post-conversion activity
Sales targets	10 branches × 3 months = 30	target, achieved, variance	Operational context
Campaign history	Lead-level	campaign_type, contacts, response	Candidate process/engagement X
Statistical validation	Summary sheet	Supplementary tests incl. response and contacts	Cross-check

2.5 Data version check

2.5 Data lineage and version reconciliation

- **Phase 1** : used an older version of the dataset.
- **Phases 2–5**: used the final version of the dataset.
- The **customer data stayed the same**, but the **lead conversion results changed**.
- To avoid confusion, the report clearly mentions this difference.
- **All conversion-related numbers in the final report are taken from the final dataset.**

Item	Phase 1 EDA (draft data)	Final workbook (Phases 2–5)	Consequence
Converted leads	626 (62.6%)	653 (65.3%)	Baseline uses 65.3%
Best lead source	Referral 65.1%; Walk-in lowest 59.2%	Digital 67.8%, Campaign 66.0%, Referral 64.8%, Walk-in 63.0%; differences not significant (p = 0.716)	Source ranking in EDA is superseded
RM conversion extremes	RM013 78.0% (best); RM014 41.7% (worst)	RM023 85.0% (best); RM019 47.1% (worst); RM014 = 75.0%	Individual RM rankings are unstable and unreliable at n ≈ 33 leads per RM
RM experience vs conversion	Visually inconsistent; 'no relationship'	Logistic OR 1.048 per year, p = 0.006 (adjusted)	Adjusted evidence supersedes the visual impression
CASA accounts	626	653 (one per converted lead)	Accounts exist only for converted leads

3. Define Phase

The Define phase turns the business problem into a clear process problem with boundaries, stakeholders and one main CTQ. Its outputs are the project charter, VOC/VOB, CTQ definition, defect definition, SIPOC and scope.

3.1 Problem and goal statements

Problem statement. The bank holds customer, lead, RM, branch, campaign, account, transaction and sales-target data, but lacks an single analytical process to measure CASA conversion, identify factors supported by the data, prioritise RM effort and monitor acquisition performance in a repeatable monitoring system.

Goal statement. Establish a measurable and repeatable CASA lead-management framework in which conversion is baselined, possible X variables (possible factors) are statistically evaluated, conversion probabilities for each lead are generated as decision support, and process performance is monitored using clear measures and named owners.

Target statement

No numeric improvement target (for example 'increase conversion by 5 points') is set, because no validated post-improvement period or controlled intervention exists. A target should be set only after the pilot in Section 6.8 supplies a real effect estimate.

3.2 Project charter

Charter element	Definition
Project title	Lean Six Sigma retail-banking analytics for CASA acquisition, lead prioritization and branch performance
Business process	Lead generation → RM assignment → contact and follow-up → CASA application / account opening → conversion recorded → monitoring
Primary Y / CTQ	CASA lead conversion rate
Possible X variables	Customer: age, income, salary-account status, credit score. Lead: source, month, campaign response, contacts. RM: experience. Branch context
Population	1,000 simulated leads linked one-to-one to 1,000 customer records
Stakeholders	Business/Bank leadership, Branch Management, Relationship Managers, Sales/Marketing, Analytics/Data Science, Risk/Governance
In scope	Conversion analytics, possible-factor analysis, lead-prioritization prototype, branch/campaign monitoring, Power BI control design
Out of scope	Deployment on real customers, automated approval/rejection, regulatory model validation, causal proof, real ROI, real before/after improvement
Time window	January – March 2026 (simulated)

3.3 Voice of Customer / Business to CTQ

Voice / stakeholder	Need	CTQ or measure
Prospective customer	Relevant, efficient engagement	Appropriate lead handling; future turnaround-time (TAT) and experience measures
Relationship Manager	Know which leads deserve attention	Conversion probability and priority band
Branch management	Understand acquisition performance	Branch-level conversion rate and lead volume

Voice / stakeholder	Need	CTQ or measure
Bank / business	More effective acquisition	Eligible-lead conversion rate
Analytics / oversight	Reliable, auditable decisions	Data quality, model performance, monitoring, human oversight

3.4 CTQ requirement and operational definition

CTQ component	Operational definition
CTQ	CASA lead conversion rate
Unit	One lead
Numerator	Leads with converted = 1
Denominator (current)	All 1,000 simulated leads. In production, use eligible/qualified leads after a validated eligibility rule
Formula	Conversion rate = converted leads ÷ eligible leads × 100
Baseline	65.3% (653/1,000)
Direction	Higher is better within the defined population
Production requirement	Validated eligibility definition and a fixed conversion window (for example, account opened within 30 days of lead creation)

3.5 Defect definition and the sigma-level decision

In Six Sigma a defect is a failure to meet a stated a customer or business requirement, counted against a defined number of opportunities. A non-converted lead is **not automatically a defect**: non-conversion can reflect customer choice, eligibility, timing or product fit. Because the project has no validated eligibility rule, clear requirement, or defined number of opportunities, **no process-sigma level or DPMO is claimed**. Reporting one would give a misleading exact number. The 65.3% figure is reported as a business baseline.

3.6 SIPOC

Suppliers	Inputs	Process	Outputs	Customers
Lead sources Customer data RM system Campaign system Branch system	Customer attributes Lead details RM assignment Campaign / contact data	1. Generate lead 2. Assign RM 3. Review / validate 4. Contact and follow up 5. CASA application 6. Account opening 7. Record conversion	Converted / non-converted status CASA account KPI data Analytical priority band	Customer RM Branch Manager Sales / Marketing Regional and senior management

3.7 High-level as-is process

1. A lead is generated and captured.
2. The lead is assigned to an RM.
3. The RM reviews available customer and lead information.
4. The customer is contacted and follow-up occurs.
5. Interest, eligibility and product fit are assessed.

6. A CASA application / account opening occurs where applicable.
7. The conversion outcome is recorded.
8. Performance is aggregated for management reporting.

The database relationships support the analytics but are not themselves a process flow map. Reliable time stamps for each step are missing, so time spent waiting or completing the process cannot be measured. Therefore, no time reduction is claimed.

3.8 Scope and boundaries

Boundary	Definition
Start	Lead enters the defined analytical population
End	Conversion status recorded for the observation window
Geography	Synthetic multi-branch retail-banking environment
Analytical scope	Descriptive statistics, statistical tests, logistic regression, ML prioritization, control design

3.9 Define tollgate

- Business problem and rationale documented.
- Primary CTQ selected, operationally defined and measurable.
- VOC/VOB translated into measures; SIPOC and boundaries established.
- Defect definition qualified; sigma/DPMO intentionally deferred with reasons.

4. Measure Phase

The Measure phase sets the current baseline for the CTQ, records how the data were collected and checked, and groups results to see where performance differs. Since the data are digital records rather than physical measurements, the ‘measurement check’ takes the form of a data-quality and coding check; a Gage R&R is not suitable here, so none is claimed.

4.1 Data collection and measurement plan

Measure	Source	Planned frequency	Owner
Eligible-lead conversion rate	Lead system	Monthly	Sales / Branch Manager
Lead source, response, contacts	Lead / campaign system	Weekly / monthly	Sales / Marketing
RM assignment and follow-up	CRM / lead system	Weekly	RM Manager
Branch leads and conversions	Branch KPI system	Monthly	Branch Manager
Model inputs	Customer / lead / RM systems	Before each scoring run	Analytics / Data
Model outcomes	Scoring log + lead outcomes	Monthly / quarterly	Analytics

4.2 Baseline CTQ

Indicator	Result	Interpretation
Total leads	1,000	Final simulated baseline population
Converted leads	653	65.3% of leads
Non-converted leads	347	34.7% of leads
CASA conversion rate	65.3%	Main baseline CTQ; 95% Wilson CI 62.3% – 68.2%
CASA accounts	653	One account per converted lead in the simulation

The the two outcome groups are not extremely unbalanced (65.3 : 34.7), so the outcome is not severely imbalanced. It matters, however, for evaluating classifiers: a model that predicts ‘converted’ for every lead would already be right 65.3% of the time, so accuracy alone is a weak measure (Section 6.5).

4.3 Monthly grouping and stability

Month	Leads	Converted	Non-converted	Conversion rate	95% Wilson CI
Jan-2026	347	225	122	64.84%	59.7% – 69.7%
Feb-2026	337	212	125	62.91%	57.6% – 67.9%
Mar-2026	316	216	100	68.35%	63.0% – 73.2%

A chi-square test of homogeneity across the three months gives $\chi^2 = 2.18$ (df = 2, **p = 0.336**), so there is no statistical evidence that conversion differs between months; all three intervals overlap the overall 65.3%. Three observations are far too few to estimate stable control limits. The chart below applies p-chart arithmetic for illustration only.

Figure 2. Illustrative p-chart. Limits use $\bar{p} = 0.653$ and each month's n ($UCL/LCL = \bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$). All points lie inside the limits; three points do not constitute a validated control chart.

4.4 Stratification by lead source

Source	Leads	Converted	Conversion rate	95% Wilson CI
Digital	233	158	67.8%	61.6% – 73.5%
Campaign	244	161	66.0%	59.8% – 71.6%
Referral	261	169	64.7%	58.8% – 70.3%
Walk-in	262	165	63.0%	57.0% – 68.6%

The 4.8-point difference between Digital and Walk-in is well inside random variation: the confidence intervals overlap almost completely, consistent with the statistical test in Section 5.2 ($p = 0.716$). Ranking sources by point estimates alone would therefore be misleading.

4.5 Customer-level descriptive baseline

Variable	Mean	Median	Std dev	Min	Max
Age (years)	40.80	41	11.62	21	60
Annual income	114,953.70	114,663	49,978.55	30,075	199,989
Credit score	723.18	721	71.98	600	850

The histograms from the exploratory phase are broad and roughly flat across their ranges, which is typical of simulated data and explains why normality tests reject in Section 5.2. Real customer data would usually show right-skewed income and clustered credit scores. The RM population ($n = 30$) has mean experience 7.8 years (range 1–14) and mean monthly target 47; the branch population ($n = 10$) has mean monthly target 500.

4.6 Data-quality and measurement-system audit

Check	Observation	Implication
Missing values	None in any explored table	Adequate for the method demonstration; production requires automated completeness checks
Duplicates	None in customers, branches, accounts	Adequate; enforce primary-key uniqueness in production
Referential integrity	Leads link to customers and RMs; accounts to customers; transactions to accounts	Must be validated before every scoring or reporting run
Coding of Y	converted $\in \{0, 1\}$; accounts = 653 = converted leads	Consistent; conversion window still to be specified
Eligibility	No eligibility flag	Blocks formal sigma / DPMO
Geography conflict	Branch table's 'state' contradicts 'city' (for example Bengaluru tagged TS in some rows and KA in others; Mumbai tagged both KA and TS). The column was dropped in EDA	Synthetic-data artefact; in a real system this would be a master-data defect
Target scale mismatch	Branch monthly targets sum to 5,000 per month, but the 30 RM monthly targets sum to only 1,410	Targets are not reconcilable, so target gap is not treated as a defect
Post-conversion tables	Accounts and transactions exist only for converted leads	Transaction behavior cannot be compared between converters and non-converters, and would leak the outcome if used as a predictor
Data version	Draft (EDA) vs final workbook (Phases 2–5)	Section 2.5; re-run EDA on final data

4.7 Operational context: targets versus achievement

The exploratory sales-target table (10 branches × 3 months) shows mean target 500 against mean achieved 218.9 (mean variance –281.1), with a maximum variance of only +39 (achievement above target is rare). This gap is large, but because the target scale is not yet checked and inconsistent with the RM-level targets (Section 4.6), it is retained as **business context**, not as a defect measure. A branch-level analysis needs SQL to join of leads, accounts and targets on a common definition.

4.8 Measure tollgate

- Primary CTQ baseline established with an uncertainty interval: 65.3% (62.3%–68.2%).
- Measurement definitions, data-collection plan and grouping variables documented.
- Data-quality audit completed; limitations recorded; formal sigma / DPMO deferred.

5. Analyze Phase

The Analyze phase asks which inputs are related to the CTQ and how strong those relationships are. The analysis was done in steps: (i) single-factor tests choose between test methods based on the variable type and data distribution; (ii) logistic regression checks the variables together and shows which relationships remain after accounting for the others; (iii) model checks are used to see whether the model is suitable.

Important wording rule

A statistically significant coefficient or test identifies an **association** or **predictive contribution**. It does not prove a causal root cause. Because the data are simulated and no real process observation or experiment was done, every finding below is a 'factor supported by the data', not a proven root cause.

5.1 Statistical analysis plan

Variable type	Comparison	Test used	Why
Categorical X vs binary Y	Independence	Pearson chi-square test (Yates correction for 2x2 tables)	Cross-tabulation counts; expected counts were adequate (all ≥ 5)
Continuous X vs binary Y	Difference in distributions	Shapiro-Wilk, then Wilcoxon rank-sum (Mann-Whitney)	The data were not normally distributed in either group, so the Wilcoxon test was used.
Several X jointly	Relationship after controlling for other variables	Binary logistic regression (logit link)	Binary outcome; odds ratios are easy to interpret; also supports model tests
Model checks	Overlap between variables, model form and fit	VIF/GVIF, Box-Tidwell, deviance, AIC	Check model assumptions before interpreting the results

5.2 Univariate statistical tests

Factor	Test statistic	p-value	Group summary	Decision ($\alpha = 0.05$)
Lead source	$\chi^2 = 1.357$, df = 3	0.7156	Conversion 63.0% – 67.8% across sources	Not significant
Occupation	$\chi^2 = 0.189$, df = 3	0.9793	Business 65.9%, Engineer 64.2%, Student 65.3%, Teacher 65.8%	Not significant
RM (30 levels)	$\chi^2 = 31.711$, df = 29	0.3327	RM rates span 47.1% – 85.0% at $n \approx 33$ each	Not significant
Salary account	$\chi^2 = 19.325$, df = 1	1.1×10^{-5}	Yes 72.3% (347/480) vs No 58.8% (306/520)	Significant
Annual income	W = 91,413	4.8×10^{-7}	Median 124,248 (converted) vs 97,582; means 120,765 vs 104,018	Significant
Credit score	W = 96,639	1.3×10^{-4}	Median 729 vs 710; means ≈ 730 vs ≈ 711	Significant
Campaign response	$\chi^2 = 60.499$, df = 1	7.4×10^{-15}	Responded 77.3% (379/490) vs not 53.7% (274/510)	Significant (process variable)
Contacts per lead	U $\approx 93,608.5$	3.8×10^{-6}	Mean 3.17 (converted) vs 2.73	Significant (process variable)

Normality checks. Shapiro-Wilk rejected normality for income (W = 0.948, $p = 9.6 \times 10^{-10}$ non-converted; W = 0.947, $p = 1.3 \times 10^{-14}$ converted) and for credit score (W = 0.962, $p = 7.5 \times 10^{-8}$; W = 0.950, $p = 5.0 \times 10^{-14}$). With n

≈ 350–650 per group the test is very sensitive and the flat distributions are the cause, but the conclusion (use rank-based tests) is the safe one either way.

Why RM identity is not significant but RM experience is. The RM chi-square spends 29 degrees of freedom on differences among 30 named RMs, each with only about 33 leads, so the test has limited power and the observed 47%–85% range can occur because of random variation. RM experience is one continuous variable with 1 degree of freedom and therefore has more ability to detect a small overall relationship. These two results are not contradictory, but RM ranking tables should not be used for performance management with this sample size; a funnel plot with binomial limits is a better way to show RM rates (Section 7).

5.3 Logistic regression with all variables together

A binary logistic model was fitted to all 1,000 leads:

$$\text{logit } P(\text{converted} = 1) = \beta_0 + \beta_1 \cdot \text{Age} + \beta_2 \cdot \text{Gender} + \beta_3 \cdot \text{Occupation} + \beta_4 \cdot \text{Income} + \beta_5 \cdot \text{SalaryAccount} + \beta_6 \cdot \text{CreditScore} + \beta_7 \cdot \text{Source} + \beta_8 \cdot \text{RMExperience}$$

The reference groups are Female, Business, salary account = No and source = Campaign. Odds ratios for continuous variables are shown using practical units (10,000 currency units of income, per year of age and RM experience) so that they are interpretable; the full one-unit coefficients are in Appendix B.

Table 5.3. Logistic regression results after adjustment (n = 1,000; 987 residual df)

Predictor	β (scaled)	SE	p-value	Odds ratio	95% Wald CI
Age (per year)	0.0277	0.0061	5.7×10^{-6}	1.028	1.016 – 1.041
Annual income (per 10,000)	0.0660	0.0142	3.2×10^{-6}	1.068	1.039 – 1.098
Salary account = Yes	0.6898	0.1423	1.3×10^{-6}	1.993	1.508 – 2.635
Credit score (per 10 points)	0.0431	0.0098	1.2×10^{-5}	1.044	1.024 – 1.064
RM experience (per year)	0.0464	0.0169	0.0061	1.047	1.013 – 1.083
Gender = Male	0.0498	0.1401	0.722	1.051	0.799 – 1.383
Occupation = Engineer	-0.1232	0.1924	0.522	0.884	0.606 – 1.289
Occupation = Student	-0.0769	0.1908	0.687	0.926	0.637 – 1.346
Occupation = Teacher	-0.1068	0.2001	0.594	0.899	0.607 – 1.330
Source = Digital	0.0357	0.2044	0.861	1.036	0.694 – 1.547
Source = Referral	-0.0292	0.1975	0.883	0.971	0.659 – 1.430
Source = Walk-in	-0.1864	0.1960	0.342	0.830	0.565 – 1.219

Figure 3. Odds ratios after adjustment with 95% confidence intervals. Intervals that exclude 1 (dashed line) are statistically significant.

Interpretation

- **Salary account** is the strongest single relationship: keeping the other variables the same, the the estimated odds of conversion are about twice as high (OR 1.99; CI 1.51–2.64). This means the odds are about twice as high; it does not mean the probability doubles.
- **Age, income and credit score** each have small effects that add up across larger changes. For example, ten extra years of age multiplies the odds by about 1.32; a 10,000 increase in income by 1.07; a 10-point rise in credit score by 1.04.

- **RM experience** shows a small positive relationship (OR ≈ 1.05 per year; ≈ 1.26 per five years). It is the smallest of the five significant terms ($p = 0.006$) and should be treated as preliminary, particularly after the multiple-testing caution in Section 5.6.
- **Gender, occupation and lead source** do not show a statistically significant relationship after adjustment; all their confidence intervals include 1.

5.4 Model fit and checks

Diagnostic	Value	Interpretation
Null deviance / residual deviance	1,291.1 (999 df) / 1,197.7 (987 df)	Deviance falls by 93.4 when the 12 predictor terms are added
Likelihood-ratio test against the baseline model	$\chi^2 = 93.4$, $df = 12$, $p \approx 1 \times 10^{-14}$	The predictors jointly improve on the intercept-only model
AIC	1,223.7	Used to compare different model versions
McFadden pseudo- R^2	0.072	Low; predictors explain only a small share of outcome uncertainty (values of 0.2–0.4 are considered excellent)
Nagelkerke pseudo- R^2	0.123	Same message on a different scale
VIF / adjusted GVIF	1.006 – 1.026 (GVIF $^{1/(2df)}$ 1.003 – 1.013)	No important overlap between predictors; the coefficients are reasonably stable
Classification at cut-off 0.5 (in-sample)	Accuracy 67.9%; sensitivity 89.4% (584/653); specificity 27.4% (95/347)	Only slightly above the 65.3% all-positive baseline; the model predicts ‘converted’ for most leads
Box-Tidwell linearity	Run, but individual p-values were not printed	Linearity of the logit cannot be claimed as confirmed ; print the test table and, if needed, use splines

Table 5.4. Sequential (Type I) analysis of deviance: contribution of each term

Term added	df	Deviance	p-value
Age	1	21.06	4.4×10^{-6}
Gender	1	0.09	0.765
Occupation	3	0.36	0.947
Annual income	1	23.25	1.4×10^{-6}
Salary account	1	20.78	5.2×10^{-6}
Credit score	1	18.89	1.4×10^{-5}
Lead source	3	1.37	0.714
RM experience	1	7.61	0.0058

The sequential results match the coefficient results: age, income, salary account, credit score and RM experience each improve model fit; gender, occupation and source do not. Together the diagnostics tell a consistent story: **there are statistically supported factors, but each is weak, so it is still hard to predict conversion for one lead.** That is precisely what the ML stage later confirms.

5.5 Supplementary sales-process variables: campaign response and contacts

The final workbook's validation sheet adds two engagement variables to the model: campaign response (Responded vs Ignored: coefficient 1.176, $p = 3.6 \times 10^{-14}$, OR 3.24) and number of contacts (coefficient 0.142, $p = 0.008$, OR 1.15 per contact). These are reported separately rather than merged into the core model.

Why these two are not treated as levers

Response and contact count are generated **during** the sales process, and may be consequences of a customer's interest rather than independent factors that cause conversion (the outcome may affect the variable), or may be recorded after the outcome is known (information leakage). Treating more contact as a way to increase conversion would be unsafe without a controlled trial.

They are therefore retained as **monitoring metrics**, and were left out of the ML input variables used for prioritization (Section 6.3).

5.6 Multiple testing, dependence and other statistical cautions

- **Multiple testing.** About eight statistical tests were run. A Bonferroni cutoff of $0.05/8 = 0.00625$ keeps salary account, income, credit score, response and contacts clearly significant, while the RM-experience coefficient ($p = 0.0061$) is right at the cutoff.
- **Independence.** Several leads belong to the same RM and branch. The standard logistic model treats them as independent, which can make the standard errors look smaller than they may be for RM-level variables such as experience. A mixed-effects logistic model with an RM-level random effect is the recommended refinement.
- **Statistical vs business importance.** With $n = 1,000$, small effects can still get small p-values. The pseudo- R^2 values show that, together, the variables explain only a small part of the variation.
- **Code note.** The R script's income Shapiro-Wilk test matches Customers to Leads by row order. The tables happen to align here, but matching records by customer_id (as used for the Wilcoxon tests) is safer.

5.7 Possible factors and supporting evidence

Category	Candidate cause	Evidence in this project	Status
Customer	Salary-account holder	$\chi^2 p < 0.001$; adj. OR 1.99	Supported
Customer	Higher income / credit score	Wilcoxon $p < 0.001$; logistic $p < 0.001$	Supported
Customer	Older age	Logistic $p < 0.001$	Supported
Customer	Gender, occupation	$p = 0.72$; $p = 0.98$	Not supported
Lead / channel	Lead source	$\chi^2 p = 0.716$; logistic $p > 0.34$	Not supported
People / RM	RM experience	Logistic $p = 0.006$	Supported, preliminary
People / RM	Which RM is assigned	$\chi^2 p = 0.333$	Not supported (low power)
Process / campaign	Campaign response, contacts	χ^2 and logistic $p < 0.01$	Supported association; not independent of the outcome
Branch / environment	Branch effect, RM workload / target pressure	Not tested with a statistical test	Gap
Data / measurement	Undefined eligibility and conversion window	Data-quality audit	Gap

5.8 Main factors supported by the analysis

X variable	Evidence	Status for DMAIC
Salary-account status	Chi-square and logistic, both $p < 0.001$	Candidate critical X
Annual income	Wilcoxon and logistic, $p < 0.001$	Candidate critical X
Credit score	Wilcoxon and logistic, $p < 0.001$	Candidate critical X
Age	Logistic $p < 0.001$	Candidate critical X
RM experience	Logistic $p = 0.006$	Possible X (preliminary)
Campaign response, contacts	$p < 0.01$ in supplementary tests	Monitoring variables (process, not independent of the outcome)
Lead source, occupation, gender, RM identity	$p > 0.05$	Not supported in current data

Decision. The results support a lead-prioritization approach that uses the five customer and RM variables (age, income, salary account, credit score, RM experience) to estimate each lead's conversion probability and to rank leads for RM attention. Because the effects are weak and not proof of cause and effect, the intervention is used only to help prioritize RM effort, not as a guarantee of higher conversion.

5.9 Analyze tollgate

- Univariate and multivariable tests completed with correct method selection.
- Assumptions checked (VIF); linearity check flagged as incomplete.
- Possible X variables identified; cause-and-effect testing is left for a real pilot.

6. Improve Phase

The Improve phase develops and chooses a solution to the problems found in Analyze, while checking the main risks. The Analyze results show small but useful patterns across several customer characteristics; that is a good use case for a scoring model that combines them, so the improvement concept is **analytical lead prioritization**: estimate each lead's conversion probability and use it to guide RM judgement, not replace it.

6.1 Solution options considered

Option	Description	Benefit	Effort	Main risk	Decision
1. Current approach	RMs choose leads by experience and intuition	None	None	Continued inconsistency	Rejected
2. Rule-based scorecard	Simple rules on salary account, income, credit score	Transparent, fast	Low	Simple but rough ranking	Keep as a backup benchmark
3. Logistic score	Probabilities from the Phase 2 model	Interpretable, easy to explain	Low	Modest AUC	Compare against Option 4 (see 6.5)
4. Combined ML models	Tuned Random Forest / XGBoost / LightGBM	Can capture complex patterns and interactions	Medium	Harder to explain; overfitting; AUC $\approx$ 0.60	Chosen as the prototype
5. RM reallocation / coaching	Move leads or train RMs based on RM ranking	Potentially large	High	RM identity not significant; rankings unstable	Deferred

These ratings are analyst judgement, not measured results.

6.2 Model setup

Element	Specification
Target	converted (1 = converted, 0 = not); 653 / 347
Predictors (5)	age, annual_income, salary_account (one-hot), credit_score, RM experience_years. Chosen from the statistically supported Xs in Analyze
Left out on purpose	Campaign response and contacts (post-outcome / not independent of the outcome risk); gender, occupation and lead source (not supported)
Train / test split	80% / 20% grouped by outcome, random_state = 123 $\rightarrow$ 800 training, 200 test (131 converted, 69 not)
Model pipeline	ColumnTransformer (one-hot encoding of salary account; numeric variables used as they are) $\rightarrow$ classifier, inside one Model pipeline so that preprocessing is fitted on training data only
Algorithms	Random Forest (class_weight = balanced), XGBoost, LightGBM
Tuning	RandomizedSearchCV, 50 parameter settings $\times$ 5-fold cross-validation = 250 fits per model; scoring = ROC-AUC
Selection rule	Highest test-set ROC-AUC (a measure of ranking ability that does not depend on one cut-off)

6.3 Model performance

Model	CV ROC-AUC (5-fold)	Test accuracy	Precision	Recall	F1	Test ROC-AUC
Random Forest	0.672	0.570	0.727	0.550	0.626	0.604
XGBoost	0.670	0.595	0.660	0.786	0.718	0.589

Model	CV ROC-AUC (5-fold)	Test accuracy	Precision	Recall	F1	Test ROC-AUC
LightGBM	not shown in exported output	0.615	0.667	0.824	0.737	0.573

Model	TN	FP	FN	TP	Specificity	Sensitivity	NPV
Random Forest	42	27	59	72	60.9%	55.0%	41.6%
XGBoost	16	53	28	103	23.2%	78.6%	36.4%
LightGBM	15	54	23	108	21.7%	82.4%	39.5%

Confusion matrices on the 200 test leads (positive class = converted). Specificity = $TN/(TN+FP)$; sensitivity = $TP/(TP+FN)$.

Figure 4. Test-set ROC-AUC with Hanley-McNeil 95% confidence intervals ($n = 200$). The confidence intervals overlap a lot.

6.4 Model choice

The tuned Random Forest was chosen because it achieved the highest test ROC-AUC (0.604) and the highest cross-validated AUC (0.672), which is the project's stated model-choice rule. Its selected settings were 700 trees, max depth 3, min_samples_split 5, min_samples_leaf 1 and all features considered at each split. The shallow depth (3) is itself informative: the best-performing forest was a strongly constrained to avoid overfitting one, suggesting that the data contain only simple patterns and more complex models may mainly fit random noise.

6.5 What the model results mean

Five simple points from the model results

- 1. Accuracy is below the all-positive baseline.** With 131 of 200 test leads converting, a model that always predicts 'converted' scores 65.5%. All three models score 57%–61.5% at the default 0.5 cut-off, so accuracy is misleading here; ROC-AUC, which measures ranking, is the better measure. (The class-weighted Random Forest balances recall and specificity, which lowers accuracy against a majority-class baseline.)
- 2. The ability to separate converters from non-converters is limited.** $AUC \approx 0.60$ means a randomly chosen converter gets a higher score than a randomly chosen non-converter only about 60% of the time. The 95% interval ($\approx 0.52 - 0.68$) barely clears 0.5.
- 3. The three models do not show a clear difference.** Test-AUC differences (0.604, 0.589, 0.573) are well within the ± 0.08 sampling variation of a 200-lead test set. Choosing Random Forest is a project choice, not proof that it is better.
- 4. Cross-validation can give a higher-looking score.** The CV AUC (0.672) is the best of 50 tuned candidates, so it carries tuning can make the score look better than it really is. The lower held-out figure (0.604) is the more realistic estimate, and even it rests on a single 200-lead split.
- 5. The main comparison model is missing.** The Phase 2 logistic regression was never scored on the same held-out set. A regularized logistic model may match or beat the ensembles with easier explanation; this comparison is the most valuable next experiment.

The results from Phases 2 and 3 together (pseudo- R^2 0.07; AUC 0.60–0.67; best forest depth 3) is that the available features carry **real but weak** information about conversion. The correct positioning is a **preliminary decision-support / lead-prioritization prototype**, not a model ready for live use.

6.6 Lead-priority groups

The selected model's predicted probability was computed for every lead and put into three action groups using pre-set cut-offs:

Priority	Rule (conversion probability p)	Leads	Share	Proposed RM action
High	$p \geq 0.70$	108	10.8%	Same-day contact; senior-RM attention; full follow-up cadence
Medium	$0.40 \leq p < 0.70$	595	59.5%	Standard follow-up cadence
Low	$p < 0.40$	297	29.7%	Lower-touch, digital or nurture journey; never ignored or rejected

The RM actions are proposed steps; they have not been implemented.

Figure 5. Distribution of the 1,000 scored leads across priority bands.

The highest-scoring lead (L00164) has a estimated probability of 0.82, and the top of the list is capped near 0.8, which again reflects the modest signal: no lead has a very high or near-certain probability. The purpose of the bands is to help RMs decide where to spend RM effort, **not to approve or reject customers automatically**.

Limits of the priority groups

- **Scores for the training data.** Probabilities were generated for all 1,000 leads by a model trained on 800 of them. For the training leads the scores are likely to look better than they really are; production scoring should apply only to new leads, or to predictions made without training on the same row for backtesting.
- **The cut-offs have not been calibrated.** 0.70 and 0.40 were chosen beforehand. The class-weighted forest can make the model probabilities differ from the 65% overall conversion rate, so the bands should be checked with a probability calibration curve or replaced with percentile groups, before use.
- **The bands should be checked on new data.** The dashboard view 'conversion rate by lead priority' should show a High should be above Medium, and Medium should be above Low on out-of-sample leads. If it does not, the the bands are not helping.

6.7 Risk review (FMEA-style)

Risk	Cause	Consequence	Level	Mitigation
Weak prediction	Limited signal in features	Effort misdirected	High	Use as an aid only; validate on new leads; keep RM override
Overfitting / training-data scoring	Scoring training rows	Overstated confidence	Medium	Score only new leads; nested or repeated CV
Cut-offs may not fit the data	Class weighting; arbitrary cut-offs	Wrong band sizes	Medium	Calibration curve; percentile bands
Fairness concern	Age and income are sensitive inputs	Unequal service levels	High	Compliance review; monitor band mix by age group; no automated rejection
Low-priority leads may get less attention	Low-band leads receive less effort, so convert less	The model may look useful only because RM effort changed	Medium	Randomised hold-out group; minimum-contact rule for all leads
Data changes over time	Real data differ from synthetic training data	Model performance may fall	High	PSI monitoring; scheduled retraining
Leakage	Using response / contacts / post-conversion data	Inflated apparent accuracy	Medium	Input-variable control: only pre-contact attributes

6.8 Pilot design: how to obtain real evidence

Whether prioritization really increases conversion can only be shown by a real real-world trial. A recommended design:

- **Hypothesis.** H_0 : conversion in the prioritised arm equals conversion in the business-as-usual arm; H_1 : it is higher.
- **Unit and allocation.** Randomise at RM level (to avoid mixing between arms) or at lead level with grouping by branch and source; keep a small comparison group with no model guidance.
- **Main metric.** Eligible-lead conversion within the defined window. Secondary: contacts per conversion, time to first contact, conversion by priority band.
- **Analysis.** Two-proportion z-test or grouped logistic regression, with pre-set stopping rules and no early checking.

Planning calculation (two-sided $\alpha = 0.05$, power 80%, baseline 65.3%, from this project's baseline; these are not results):

Target uplift (percentage points)	Target rate	Leads needed per arm
+3	68.3%	≈ 3,865
+5	70.3%	≈ 1,367
+7	72.3%	≈ 684
+10	75.3%	≈ 324

Small improvements need many leads in each group, so the pilot should run long enough, or across enough branches, to reach the required volume; otherwise a result showing no improvement would be hard to interpret.

6.9 Improve phase review

- Solution options generated and compared; ML prioritization selected as a prototype.
- Models compared with appropriate metrics; limitations quantified and stated.
- Risks identified; a real pilot specified. **No claim of actual improvement is made.**

7. Control Phase

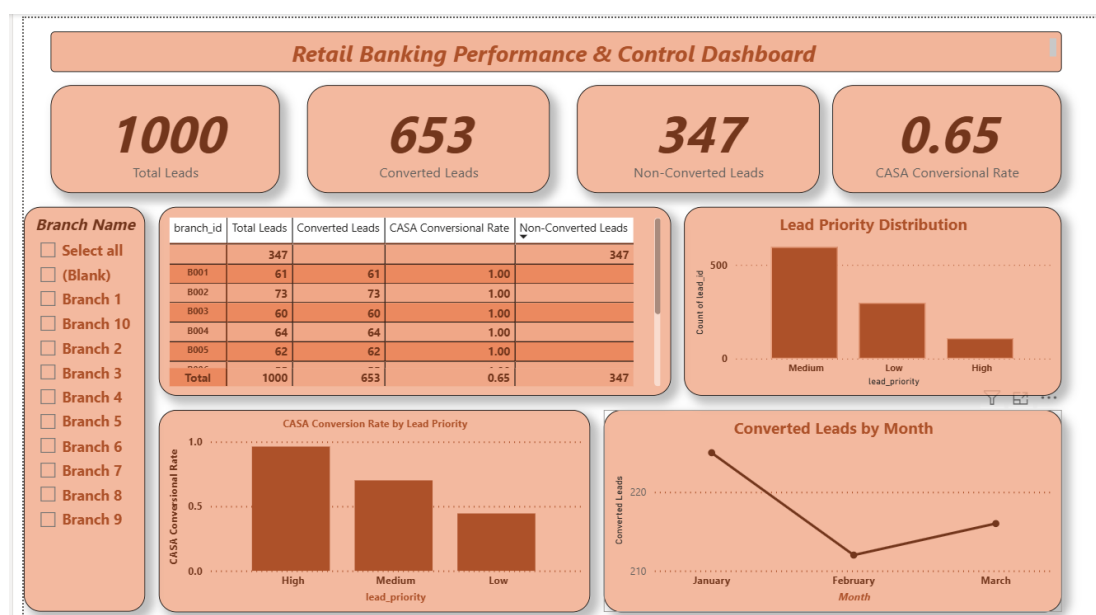
The Control phase helps keep the improved process stable: the CTQ and its key factors are monitored, owners are assigned, and a clear action is taken when performance moves outside the expected range. The project delivers three control tools: a Power BI dashboard, a control plan, and monitoring rules for the model and the data.

7.1 Power BI Retail Banking Intelligence & Control Dashboard

The delivered Power BI file (CASA_DESKSTOP_PRO.pbix) contains a single 1280 × 720 report page whose visuals use three tables (Branches, Leads and the ML priority table); the full data model could not be read from the file. The report layout shows the following components:

Component	Visual type	Content	Control purpose
KPI cards (4)	Card	Total leads; converted leads; non-converted leads; CASA conversion rate	Headline CTQ and volume at a glance
Monthly trend	Line chart	Converted leads by month (lead_month hierarchy)	Trend / shift detection
Lead-priority distribution	Clustered column	Count of leads by High / Medium / Low	Shows changes in the mix of High / Medium / Low leads
Conversion by priority	Clustered column	Conversion rate by priority band	Model check (High should be above Medium and Low)
Branch performance table	Table	Branch ID, total leads, converted leads, conversion rate	Branch comparison
Branch slicer	Slicer	Branch name	Interactive filtering
Text box	Annotation	Free-text commentary (content not extracted)	Interpretation aid

Dashboard improvements to make (a) The the DAX measures could not be fully checked because the data model is compressed, so measure logic should be documented in a data dictionary. (b) The rate measure is titled 'CASA Conversion Rate'; rename it 'CASA Conversion Rate' to fix the wording error. (c) Add a p-chart (or at least a target or average line) to the trend, a campaign-response view, and a funnel plot for branch and RM comparisons so that small-sample variation is not mistaken for a real performance difference. (d) Show the number of leads beside every rate, and label the dashboard as simulated data until real data are connected.



7.2 Control plan

Specification limits, thresholds and frequencies below are **proposed initial values** to be approved by the process owners; none were derived from real real-world history.

#	Metric (CTQ / driver)	Definition and tool	Frequency	Owner	Reaction if signal
1	CASA conversion rate (primary CTQ)	Converted ÷ eligible leads; p-chart with variable limits	Monthly (weekly when volume allows)	Sales Head / Branch Manager	Investigate within 5 working days; stratify by branch, source, RM, priority; log root cause
2	Conversion by priority band	Rate per band; expect High > Medium > Low	Monthly	Analytics	If High ≤ Medium for two months, suspend banding and review the model
3	Priority-band mix	Share High / Medium / Low	Monthly	Analytics	Large shift in mix triggers input-changes over time check
4	Branch conversion	Funnel plot with binomial limits	Monthly	Branch Manager	Outside 99.8% limits triggers a branch review; inside limits, no action
5	Campaign response and contacts per lead	Response rate; mean contacts; by campaign type	Monthly	Sales / Marketing	Review targeting and contact cadence
6	Follow-up adherence (future)	High-priority leads contacted within the SLA; needs timestamps	Weekly	RM Manager	Coach RMs; check workload allocation
7	Data quality	Missing %, duplicates, invalid values, referential integrity	Before every scoring run	Data owner	Stop scoring and fix the source if a check fails
8	Model discrimination	Rolling AUC and probability calibration on recent leads with known outcomes	Quarterly	Analytics	AUC below 0.55, or worse than the logistic benchmark, triggers retraining and review
9	Input changes over time	Population Stability Index per feature	Monthly	Analytics	PSI 0.10–0.25 watch; above 0.25 investigate / retrain
10	Fair-treatment and override	Band mix by age group; share of RM overrides	Quarterly	Risk / Compliance	Review with oversight forum; adjust features or process

7.3 Statistical process control setup

Conversion is a proportion, and the number of leads can change by period, so a **p-chart** with variable control limits is appropriate: average line $\bar{p}$ = total converted ÷ total leads; limits $\bar{p} \pm 3\sqrt{\bar{p}(1 - \bar{p})/n_i}$ for period i , truncated to $[0, 1]$. Figure 2 applied this to the three baseline months; every point lies inside its limits.

- **Phase I (establish limits).** Collect at least 20–25 rational subgroups (for example, weekly or branch-month) with stable process conditions before fixing limits; three months is insufficient.
- **Phase II (monitor).** Apply run rules, for example one point beyond 3 standard deviations, eight consecutive points on one side of the average line, or six points moving steadily in one direction. Each rule fires only on data outside the Phase I window.
- **Check for extra variation.** If points scatter more than binomial theory predicts (for example because RMs differ), use a Laney p'-chart to avoid false alarms.
- **Separate signals from noise.** An unusual signal should be investigated; normal process variation calls for process improvement, not blame.

7.4 Model and data monitoring, oversight

- **Human review.** Every priority band is advice. RMs may override, and overrides are logged and reviewed.

- **Versioning.** Store the training data snapshot, feature list, model settings, random seed and metrics with each model version; retrain on a schedule or on a trigger from the control plan.
- **Input-variable control.** Any new predictor must be available before first contact and free of information created after the outcome.
- **Documentation.** Keep the data dictionary, dashboard measure definitions and this report under version control.

7.5 Roles and responsibilities

Role	Responsibility
Business sponsor / Sales Head	Owns the CTQ target and approves the control plan and pilot
Branch Managers	Review branch performance; act on branch-level signals
RM Manager and RMs	Follow priority guidance and record outcomes accurately; log overrides
Sales / Marketing	Own campaign strategy and contact cadence
Analytics / Data Science	Maintain data pipelines, model, dashboard and monitoring; run retraining
Risk / Compliance	Model-risk review; fair-treatment oversight; approve feature set

7.6 Standard process and ongoing monitoring

- Convert the priority-band procedure into a standard operating procedure (SOP) with a one-page RM one-page guide.
- Train RMs and Branch Managers on reading the dashboard, and on the difference between random variation and a real warning signal.
- Hold a monthly performance review and a quarterly model review; record actions and closure dates.
- Replicate the framework to other products (for example, term deposits or cards) only after the CASA pilot yields validated results.

7.7 Control tollgate

- Dashboard, control plan, SPC design, model/data monitoring and reaction plan documented.
- Ownership and review schedule defined. **The controls are designed but have not been tested on live data.**

8. Overall Discussion

8.1 Evidence across the phases

Step	Evidence	What it supports	What it does not support
Define	CTQ, scope, defect definition	A precise, measurable problem	A sigma level
Measure	65.3% (62.3–68.2%); months homogeneous	A stable-looking baseline	Statistical control (n = 3 periods)
Analyze	Five possible X variables; pseudo-R ² 0.07	Genuine but weak associations	Causation; large effect sizes
Improve	Test AUC 0.60; three bands	A ranking aid worth piloting	Production readiness; realised uplift
Control	Dashboard + control plan	A monitoring system	Proof of long-term improvement

Each phase only makes claims that the available evidence supports, and the report states the limits clearly. That consistency, more than any single number, is what makes the project a sound Six Sigma analysis.

8.2 Limitations that can affect the results

#	Limitation	Effect	Remedy
1	Synthetic data; flat distributions; generated relationships	Results show method, not real behavior	Replicate on real, governed data
2	Three-month window; no timestamps	No trend, seasonality, cycle-time or SPC baseline	Collect ≥ 20–25 subgroups; add event timestamps
3	Observational data	Association only; possible confounding	Randomised pilot
4	Undefined eligibility and conversion window	No defect, DPMO or sigma level	Define eligible leads and window
5	Draft vs final data versions	Some EDA numbers are superseded	Re-run EDA on the final workbook
6	Small held-out set (n = 200); one split	Wide AUC interval; unstable model ranking	Repeated / nested CV; larger validation
7	Tuning optimism; in-sample band assignment	Overstated performance	Score only new leads; nested CV
8	No logistic benchmark on the same test set	Value of ensembles unproven	Compare regularized logistic model
9	Non-independence (leads within RMs / branches)	Standard errors may be too small	Mixed-effects logistic regression
10	Box-Tidwell result not reported	Linearity of the logit unverified	Report tests; use splines if needed
11	Endogenous sales-process variables (response, contacts)	Not usable as levers or ML features	Experimental validation
12	Branch effects and RM workload untested	Possible confounders omitted	Add branch/RM multilevel terms
13	Age, income as inputs to a customer-facing ranking	Fair-treatment risk	Compliance review and monitoring

8.3 Statistical significance vs business importance

All five headline Xs are statistically significant, yet the model's explanatory power is small. In practice this means the bank cannot reliably say in advance that a particular lead will or will not convert; it can only say that some

leads are somewhat more likely than others. The practical approach is to: use scores only to guide some RM effort, avoid over-reliance on the model, and test the idea in a real pilot.

8.4 Next-step plan

Priority	Action	Purpose	Indicative horizon
1	Re-run EDA, tests, ML and dashboard on the single final workbook; fix the state/city and target-scale issues	One consistent data version	Immediate
2	Score the logistic model on the held-out set; add probability calibration curve, precision-recall curve, permutation importance or SHAP	Test whether ML adds value; explain the model	1–2 weeks
3	Repeated grouped CV and nested CV; report AUC with CI	Reliable model comparison	1–2 weeks
4	Mixed-effects logistic regression (RM and branch random intercepts); Box-Tidwell table or splines	Correct dependence; verify linearity	2–3 weeks
5	Define eligibility and conversion window; build SQL integration of leads, accounts, targets	Enable sigma/DPMO and true branch analysis	3–4 weeks
6	Harden the dashboard: p-chart, funnel plots, n beside each rate, data dictionary	Operational control	2 weeks
7	Run the randomized pilot (Section 6.8)	Real, causal evidence of uplift	3–6 months

9. Conclusion

This project follows the full DMAIC approach to a CASA acquisition problem. It defines a single measurable CTQ, establishes a 65.3% baseline with an confidence interval, shows through appropriate statistical tests and a multivariable logistic model that customer financial characteristics (salary-account status, income, credit score, age) and RM experience are statistically associated with conversion while lead source, occupation, gender and RM identity are not, and uses these results to build a easy-to-understand lead-prioritization prototype with a Power BI monitoring layer and a written control plan.

Its principal strengths are method-related: correct test selection, effect sizes shown along with p-values, assumption checks, an honest evaluation of the ML models, clear separation of relationship from cause and effect, and a control system that anticipates changes over time, fairness and oversight.

Its principal limitations are equally clear: simulated data, a short window, modest predictive power, and no operational experiment. The next step is the roadmap in Section 8.4, culminating in a randomized pilot that would provide the real evidence required to claim improvement.

Bottom line

A rigorous, reproducible Lean Six Sigma analytics framework and decision-support prototype: **ready for a real pilot, but not yet proven to improve conversion.**

Appendix A. Glossary

Term	Meaning
CASA	Current Account and Savings Account; low-cost deposit products
CTQ	Critical to Quality: a measurable result that matters to the customer or business
DMAIC	Define, Measure, Analyze, Improve, Control: the Six Sigma improvement process
SIPOC	Suppliers, Inputs, Process, Outputs, Customers: a simple high-level process map
VOC / VOB	Voice of Customer / Voice of Business
DPMO	Defects per million opportunities; used to calculate process sigma
RM	Relationship Manager
Odds ratio (OR)	How the odds of the outcome change when a predictor changes by one unit, while other variables stay the same
ROC-AUC	Chance that a random converted lead gets a higher score than a random non-converted lead (0.5 = chance, 1 = perfect)
McFadden R ²	1 minus residual deviance divided by null deviance; a fit measure for logistic regression
VIF / GVIF	Variance inflation factor / generalised VIF; a check for overlap between predictors
p-chart	Control chart for the proportion of units with a given result (here, converted leads)
PSI	Population Stability Index: measure of input-distribution changes over time
Wilson interval	Confidence interval for a proportion with good coverage at moderate n

Appendix B. Full Phase 2 Logistic-Regression Output (per-unit scale)

Term	Estimate	Std. error	z	p-value	Odds ratio
(Intercept)	-4.895	0.8246	-5.935	2.93×10^{-9}	0.0075
Age	0.02773	0.006111	4.537	5.69×10^{-6}	1.0281
Gender: Male	0.04981	0.1401	0.356	0.7222	1.0511
Occupation: Engineer	-0.1232	0.1924	-0.641	0.5218	0.8840
Occupation: Student	-0.07694	0.1908	-0.403	0.6867	0.9259
Occupation: Teacher	-0.1068	0.2001	-0.534	0.5935	0.8987
Annual income	6.599×10^{-6}	1.418×10^{-6}	4.655	3.24×10^{-6}	1.0000066
Salary account: Yes	0.6898	0.1423	4.848	1.25×10^{-6}	1.9934
Credit score	0.004308	0.0009843	4.377	1.20×10^{-5}	1.0043
Source: Digital	0.03569	0.2044	0.175	0.8614	1.0363
Source: Referral	-0.0292	0.1975	-0.148	0.8825	0.9712
Source: Walk-in	-0.1864	0.1960	-0.951	0.3415	0.8299
RM experience (years)	0.04639	0.01692	2.743	0.0061	1.0475

Null deviance 1,291.1 on 999 df; residual deviance 1,197.7 on 987 df; AIC 1,223.7; 4 Fisher-scoring iterations. Confusion matrix at cut-off 0.5 (in-sample): actual 0 → predicted 0 = 95, predicted 1 = 252; actual 1 → predicted 0 = 69, predicted 1 = 584.

Appendix C. Reproduction and File List

File	Content	Used for
PHASE_-_1DATA_EXPLORATION.pdf	Python EDA notebook (draft-data version)	Section 4 structure, data quality, context
PHASE_-_2_Statistical_test_and_regression_modelling.R / .docx	R tests, logistic regression, VIF, Box-Tidwell; interpretation	Section 5
PHASE_-_3_MACHINE_LEARNING_VALIDATION_AND_PREDICTION.html	Python notebook: RF, XGBoost, LightGBM, priority bands	Section 6
CASA_DESKTOP_PRO.pbix	Power BI dashboard	Section 7.1
DMAIC_01 / 02 / 03 (.docx)	Define, Measure and Analyze phase write-ups	Sections 3–5
Banking_Intelligence_Modified.xlsx (referenced, not supplied)	Final synthetic workbook, nine sheets	All phases from Measure onward
ML_Lead_Priority.xlsx (output of the notebook)	Lead ID, customer, RM, probability, priority	Dashboard priority table

Reproducibility settings: train/test split random_state = 123 (grouped, 80/20); RandomizedSearchCV random_state = 123, 50 iterations, 5-fold CV, scoring roc_auc; Random Forest class_weight = balanced. Python libraries: pandas, numpy, scikit-learn, xgboost, lightgbm. R libraries: readxl, dplyr, car. Recommendation: move the notebook out of the current absolute path (a TensorFlow site-packages directory) into a project folder with a requirements file and relative paths.

Appendix D. Resume Statements

wording that matches what was actually demonstrated,

1. Designed and executed an end-to-end Lean Six Sigma (DMAIC) analytics project on a simulated 1,000-lead retail-banking CASA dataset, defining the CTQ, and establishing a 65.3% (95% CI 62.3–68.2%) conversion baseline with data-quality audit and grouping.
2. Identified statistically supported factors related to conversion using chi-square, Wilcoxon and multivariable logistic regression (LR $\chi^2 = 93.4$, $p < 0.001$; VIF ≈ 1): salary-account status (adjusted OR ≈ 2.0), income, credit score, age and RM experience; found lead source, occupation and RM identity non-significant.
3. Built and tuned Random Forest, XGBoost and LightGBM lead-scoring models (5-fold CV AUC 0.67; held-out AUC 0.60) with a clear statement of their limits; produced High/Medium/Low priority bands as a decision-support prototype.
4. Developed a Power BI control dashboard and a Lean Six Sigma control plan (p-chart design, model-changes over time and data-quality monitoring, reaction plan) and specified a randomized pilot with sample sizes based on the planned power.
5. **One-line version:** ‘Lean Six Sigma DMAIC analytics framework for CASA lead conversion, combining hypothesis testing, logistic regression, ML lead prioritization and Power BI process control (simulated data).’